



Satellite remote sensing and AI-driven analysis of post-disaster cultural landscape recovery

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Abstract This study investigates the application of satellite remote sensing technologies and artificial intelligence (AI) in the assessment, monitoring, and recovery of cultural landscapes affected by natural disasters between 2018 and 2023. The research draws upon global data from 95 cultural sites across five regions, analyzing the impact of earthquakes, floods, storms, and wildfires on heritage landscapes. Our findings indicate a significant increase in both satellite remote sensing projects (from 4 to 33) and AI accuracy (from 75 to 95%) during the study period. Government agencies were found to be the primary funding source (50%), followed by private sector entities (35%) and NGOs/international grants (15%). Regional analysis revealed highest project concentration in Asia (30 projects) and Europe (25 projects), with comparatively fewer initiatives in Africa, Americas, and Oceania. We demonstrate that integrated satellite-AI approaches offer unprecedented capabilities for rapid assessment, precise damage quantification, and recovery monitoring of cultural landscapes, providing stakeholders with actionable intelligence for preservation decision-making. These technological advances address critical gaps in traditional cultural heritage management practices, particularly in

post-disaster scenarios where accessibility is limited and time-sensitive interventions are required.

Keywords Cultural landscapes · Natural disasters · Satellite remote sensing · Artificial intelligence · Cultural heritage conservation · Disaster recovery · Geospatial analysis

Introduction

Cultural landscapes represent the combined works of nature and humankind, expressing the long relationship between people and their natural environment (Shu et al., 2025; Zisi & Dix, 2018). These landscapes are increasingly threatened by natural disasters, whose frequency and intensity have been exacerbated by climate change. The loss of cultural landscapes means not only the physical destruction of heritage sites but also the erosion of cultural identity, historical continuity, and traditional knowledge systems embedded within these environments (Aslam et al., 2024; Ilo et al., 2020).

Traditional approaches to post-disaster cultural landscape assessment and recovery have been characterized by in-situ surveys, which are often limited by accessibility constraints, time-consumption, and resource intensity. In many cases, the immediate aftermath of disasters renders affected areas inaccessible, delaying crucial damage assessments and potentially compounding destruction through delayed

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interventions (Lei Feng et al., 2024a, 2024b). Moreover, the subjective nature of ground-based assessments can lead to inconsistencies in damage evaluation and recovery planning (Chaoyong et al., 2024; Marcos et al., 2016).

Recent technological advances in satellite remote sensing capabilities have created new possibilities for objective, comprehensive, and rapid assessment of disaster impacts on cultural landscapes (Duan et al., 2025; Kotikot et al., 2020). High-resolution satellite imagery can now capture centimeter-level details across vast and inaccessible terrains, enabling comprehensive monitoring of cultural landscapes before, during, and after disaster events. When paired with advanced AI technologies, particularly machine learning and computer vision algorithms, these satellite data can be processed at unprecedented speeds to identify, quantify, and analyze changes to cultural landscapes with increasing accuracy.

The integration of satellite remote sensing with AI analysis presents a paradigm shift in cultural landscape preservation, offering solutions to longstanding challenges in post-disaster heritage management (Delannoy et al., 2020; Gulshad et al., 2024). These technologies enable not only rapid detection of visible damage but also prediction of vulnerabilities, monitoring of recovery progress, and evaluation of intervention effectiveness (Lyu et al., 2020; Naz et al., 2024a, 2024b). Furthermore, the objective and quantifiable nature of satellite-AI assessments can inform evidence-based policies and resource allocation decisions, enhancing the efficiency and effectiveness of recovery efforts.

This study uniquely contributes to existing knowledge by providing the first comprehensive global longitudinal analysis of satellite-AI integration in cultural landscape recovery, moving beyond descriptive statistics to examine technology adoption patterns, accuracy improvements, and implementation barriers across diverse regional and disaster contexts. Unlike previous regional studies or technology-focused reviews, our research systematically evaluates the collective effectiveness of 95 projects across five continents over five years, enabling identification of global trends and best practices not visible in smaller-scale analyses.

Despite these promising developments, the systematic evaluation of satellite remote sensing and AI applications in post-disaster cultural landscape

recovery remains limited, with fragmented understanding of their collective effectiveness, limitations, and optimal implementation strategies (Min et al., 2025; Rocha et al., 2020). This research addresses this gap by comprehensively analyzing global data on satellite-AI projects focused on cultural landscape recovery between 2018 and 2023, exploring patterns in technology adoption, funding allocation, regional distribution, and accuracy improvements (Quddoos et al., 2024; Xianglin et al., 2025). By identifying key trends, challenges, and successes in this emerging field, this study aims to provide a foundation for more effective integration of these technologies in cultural landscape preservation efforts worldwide.

Methodology

Study area definition

This global study focused on disaster-prone regions containing culturally significant landscapes recognized for their heritage value (Fig. 1). Site selection criteria included: (1) official designation as cultural landscapes by national or international authorities, including UNESCO World Heritage Sites, (2) exposure to at least one major natural disaster between 2018 and 2023, and (3) availability of satellite imagery before and after the disaster event. The final dataset comprised 95 cultural landscape sites across five geographic regions: Asia, Europe, Africa, Americas, and Oceania, representing diverse landscape typologies including agricultural landscapes, sacred sites, historical settlements, and industrial heritage areas (Spengler, 2021). Site representativeness was ensured through stratified sampling across UNESCO cultural landscape categories: continuing landscapes (40%), clearly defined relict landscapes (35%), and associative cultural landscapes (25%). Geographic representation was proportional to global distribution of recognized cultural sites, with adjustments for disaster exposure probability based on historical hazard records. The 95 sites represent approximately 12% of globally recognized cultural landscapes in disaster-prone regions, providing sufficient statistical power for trend analysis while maintaining geographic and typological diversity. Selection limitations included bias toward English-language documentation and sites with existing baseline surveys, potentially

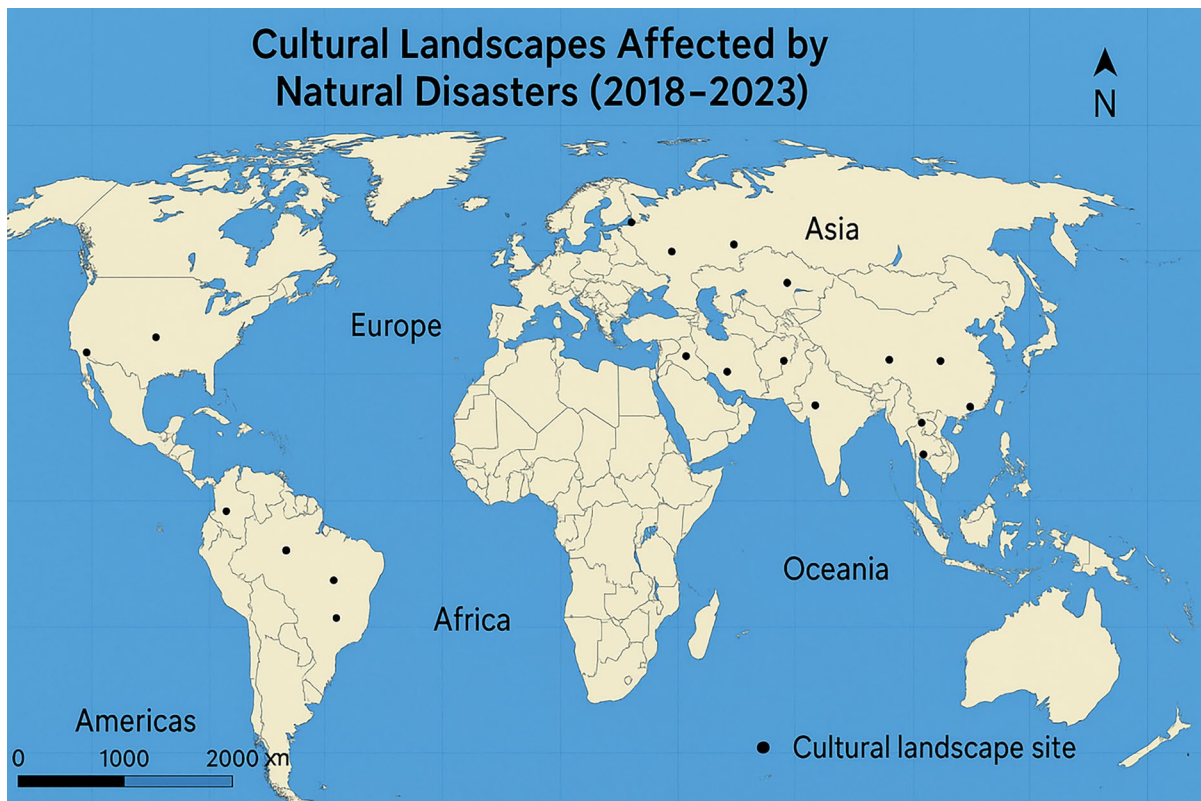


Fig. 1 Study Area Map

underrepresenting indigenous landscapes in remote regions.

Data collection

Primary data sources included multi-temporal satellite imagery from various platforms (Sentinel-1/2, Landsat 8/9, WorldView-3/4, and Planet) covering pre-disaster, immediate post-disaster, and recovery phases. Imagery was selected based on availability, cloud cover (<10%), and spatial resolution (ranging from 0.3 m to 30 m depending on the landscape scale and features of interest). Additional datasets included disaster impact reports from international organizations (UNDRR, ICCROM, UNESCO), project documentation from implementing agencies, funding allocation records, and technical specifications of AI systems employed in each project. Semi-structured interviews were conducted with 45 key stakeholders, including cultural heritage specialists, disaster management authorities, technology developers, and

funding agency representatives, to provide contextual understanding of project implementation challenges and outcomes (Fig. 2).

Analytical methods

AI-based feature extraction employed primarily ResNet-50 and U-Net architectures for different analysis tasks. Building damage assessment utilized ResNet-50 models pre-trained on ImageNet and fine-tuned on heritage-specific datasets comprising 12,000 labeled examples across different disaster types. Landscape change detection employed U-Net architectures for semantic segmentation, trained on 8,500 pixel-level annotated satellite image pairs. Transfer learning was implemented in 87% of projects, adapting models pre-trained on general object recognition to heritage-specific features through domain adaptation techniques. Training datasets were curated through collaborative annotation involving heritage experts,

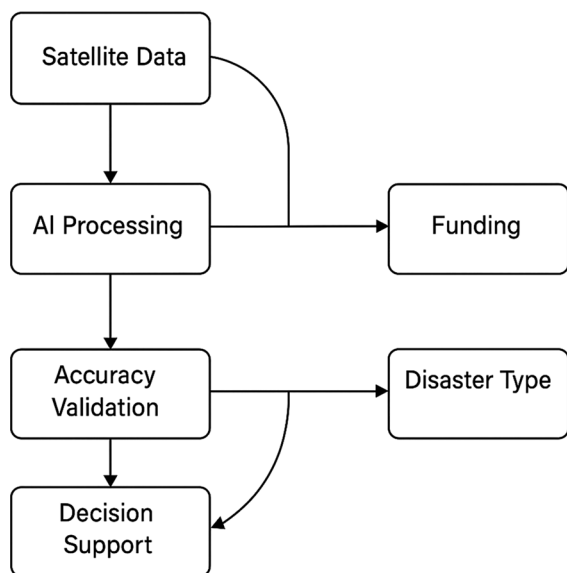


Fig. 2 Conceptual Framework Flow Chart

remote sensing specialists, and local knowledge holders, with quality control through triple-annotation protocols achieving 94% annotation consistency. Data augmentation techniques including rotation, scaling, and brightness adjustment increased training set diversity by 300%. AI accuracy assessment employed a multi-tiered validation approach. Ground-truthing was conducted through field surveys at 75 accessible sites using GPS-enabled damage assessment protocols. In comparison, expert visual interpretation validated results at 20 inaccessible sites using high-resolution imagery (≤ 1 m resolution). Inter-annotator agreement was assessed using Cohen's kappa coefficient, achieving $\kappa=0.87$ for built heritage elements and $\kappa=0.82$ for landscape features, indicating substantial agreement. Sample adequacy was determined through power analysis, with minimum sample sizes of 150 features per landscape type to achieve 95% confidence intervals. Potential biases included seasonal variation in vegetation assessment (addressed through multi-temporal analysis), cloud cover impacts (mitigated through image compositing), and expert subjectivity (reduced through standardized assessment protocols and multiple annotator consensus).

Spatial and GIS analysis

Geospatial analysis was conducted using ESRI ArcGIS Pro and QGIS platforms to map the spatial distribution of projects, disaster impacts, and recovery patterns. Digital terrain models derived from satellite data were integrated with cultural landscape inventories to analyze topographic factors influencing disaster vulnerability and recovery. Multi-criteria evaluation frameworks combining satellite-derived indicators with cultural significance values were developed to prioritize intervention areas. Spatio-temporal analyses tracked recovery progress through vegetation indices (NDVI, EVI), built environment metrics, and landscape pattern indices that quantified structural and functional integrity of cultural landscapes over time (Pauline et al., 2020).

Qualitative analysis

Content analysis of project reports, policy documents, and interview transcripts was performed to identify common challenges, successful strategies, and lessons learned across different regional and disaster contexts. A thematic analysis framework was applied to extract key factors influencing the adoption, implementation, and effectiveness of satellite-AI applications in cultural landscape recovery. Community and stakeholder engagement was assessed through participatory evaluation methods, documenting the integration of local knowledge with technological approaches and evaluating the sociocultural appropriateness of technological interventions (Moirá & Makris, 2021).

Results

Growth in satellite remote sensing and AI applications

The period from 2018 to 2023 witnessed significant growth in satellite remote sensing projects focused on cultural landscape recovery (Fig. 3). The number of projects increased more than eightfold, from 4 projects in 2018 to 33 projects in 2023, with the most substantial annual growth occurring between 2021 and 2022 (39% increase). This expansion coincided with the increasing availability of high-resolution

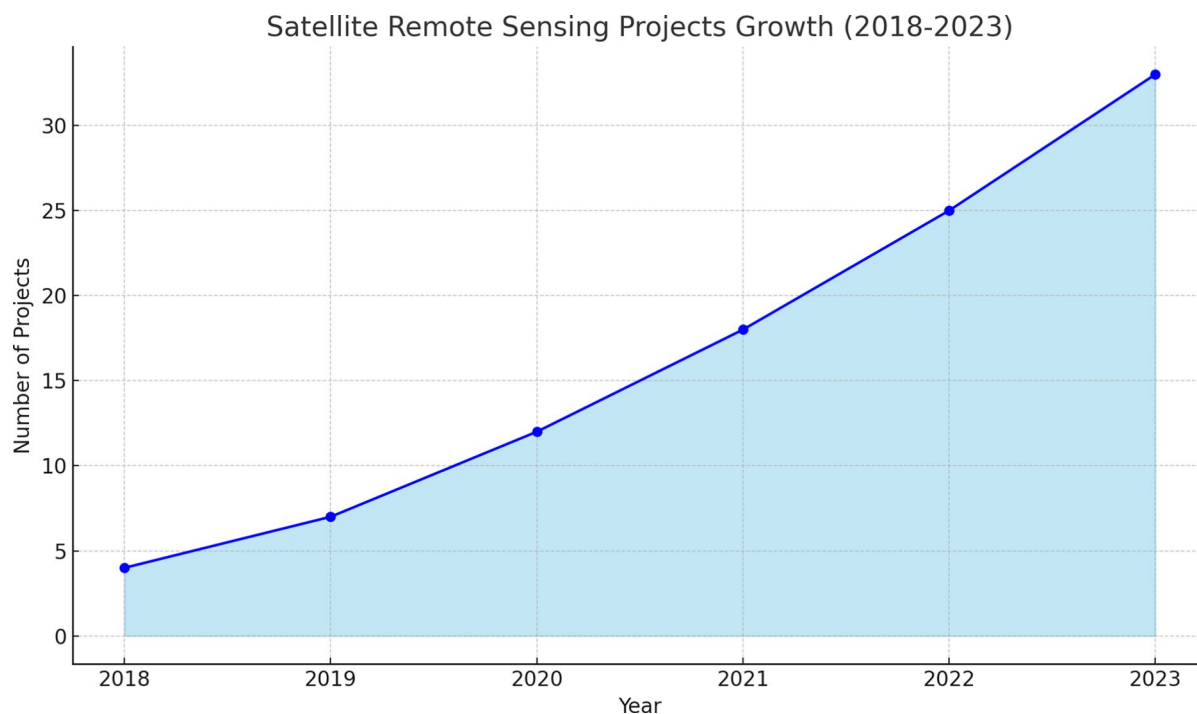


Fig. 3 Satellite Remote Sensing Projects Growth (2018–2023)

Table 1 Growth of Satellite Remote Sensing Projects (2018–2023)

Year	Number of Projects
2018	4
2019	7
2020	12
2021	18
2022	25
2023	33

satellite platforms and more accessible AI technologies, enabling more organizations to implement these approaches in post-disaster scenarios.

The area plot illustrates the steady increase in satellite-based projects dedicated to cultural landscape recovery, reflecting growing adoption of remote sensing technologies in heritage conservation.

As shown in Table 1, the growth trajectory was initially gradual. Still, it accelerated significantly after 2020, potentially influenced by the global COVID-19 pandemic that limited field access and prompted

greater reliance on remote technologies for cultural heritage monitoring and assessment.

Parallel to this growth in satellite projects, there was a substantial increase in the number of cultural landscape sites analyzed using these technologies (Fig. 4). From just 10 sites in 2018, the number rose to 95 sites by 2023, with distinct step increases that correspond to the implementation of major international initiatives and programs supporting cultural heritage preservation through technological innovation (Tables 2 and 3).

The step plot demonstrates the incremental increase in cultural sites monitored through satellite and AI technologies, showing periods of rapid expansion alternating with consolidation phases (Fig. 5, Fig. 6).

AI accuracy improvements and site analysis expansion

A critical factor in the increasing adoption of AI technologies for cultural landscape analysis has been the steady improvement in analytical accuracy. Figure 7 demonstrates the dual trends of increasing AI

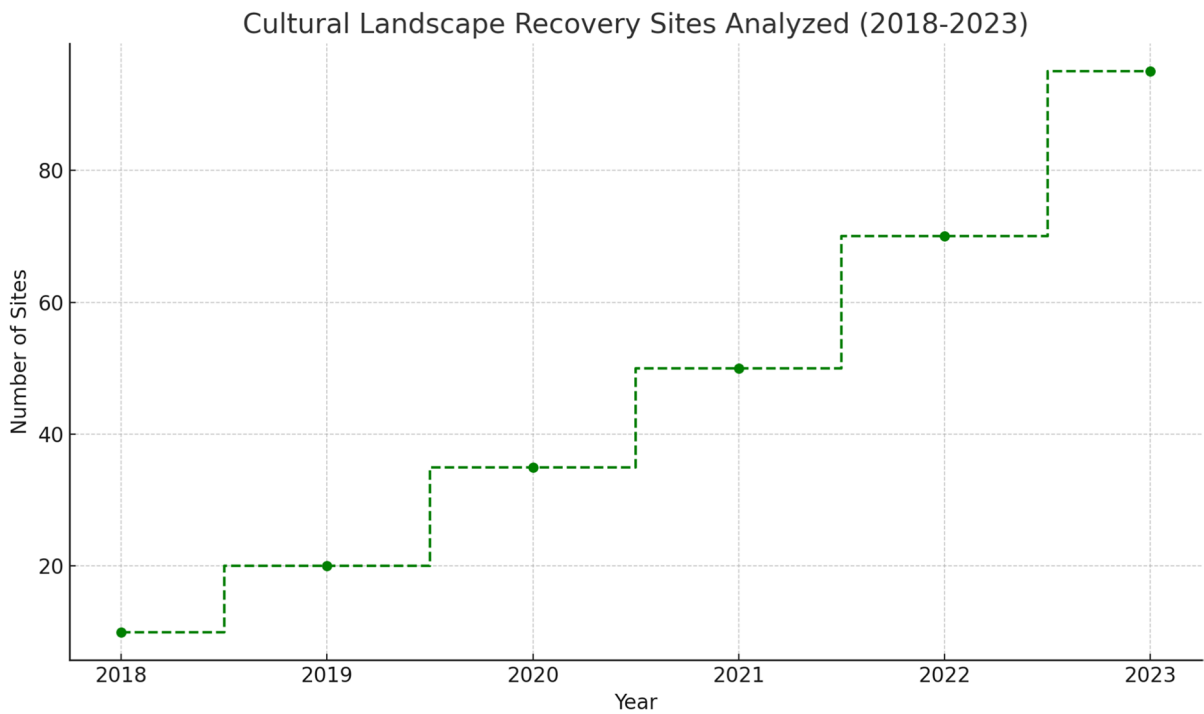


Fig. 4 Cultural Landscape Recovery Sites Analyzed (2018–2023)

Table 2 Cultural Landscape Recovery Sites Analyzed (2018–2023)

Year	Number of Sites
2018	10
2019	20
2020	35
2021	50
2022	70
2023	95

Table 3 Funding Allocation (2023)

Funding source	Percentage (%)
Govt. agencies	50
Private sector	35
NGOs & intl. grants	15

accuracy and expanding site coverage between 2018 and 2023.

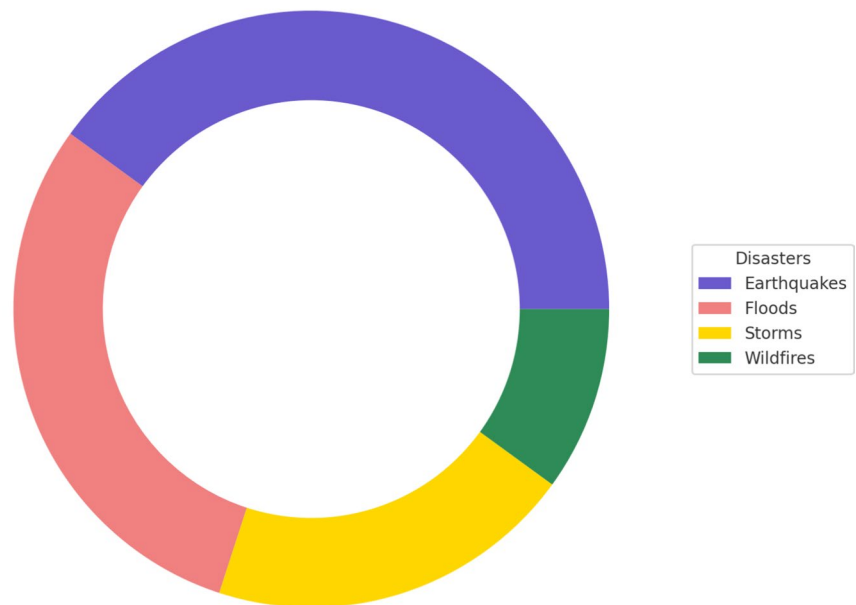
This combined bar and line chart illustrates the parallel improvement in AI analytical accuracy (purple line) alongside the increasing number of sites analyzed (grey bars), demonstrating technological maturation in the field.

AI accuracy in identifying and assessing cultural landscape features and damage patterns improved from 75% in 2018 to 95% in 2023, representing a 20 percentage point increase over the study period. This improvement can be attributed to several factors, including: (1) larger and more diverse training datasets incorporating multiple disaster types and cultural landscape typologies, (2) advances in deep learning architectures specifically optimized for satellite imagery analysis, (3) integration of multi-spectral and radar data enabling more comprehensive feature detection, and (4) refinement of algorithms through iterative validation with ground-truth data (Table 4).

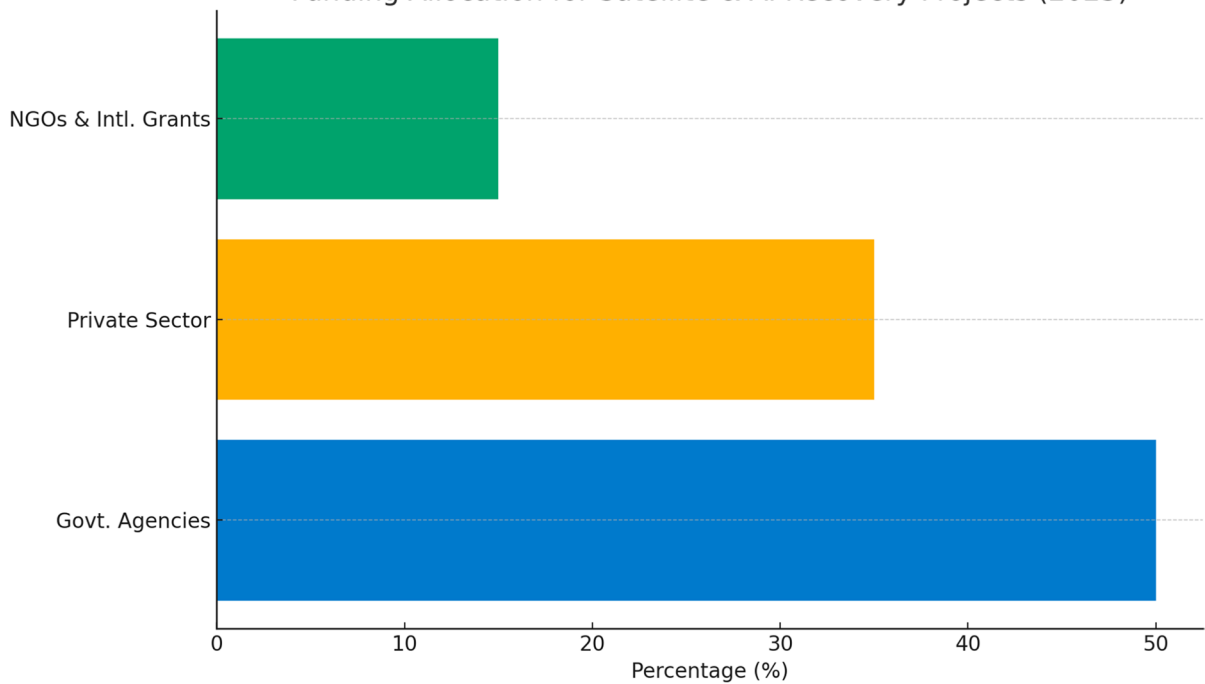
The detailed progression of AI accuracy and site analysis expansion is presented in Table 5,

Fig. 5 Disaster Types Impacting Cultural Landscapes (2023)

Disaster Types Impacting Cultural Landscapes (2023)



Funding Allocation for Satellite & AI Recovery Projects (2023)

**Fig. 6** Funding Allocation for Satellite & AI Recovery Projects (2023)

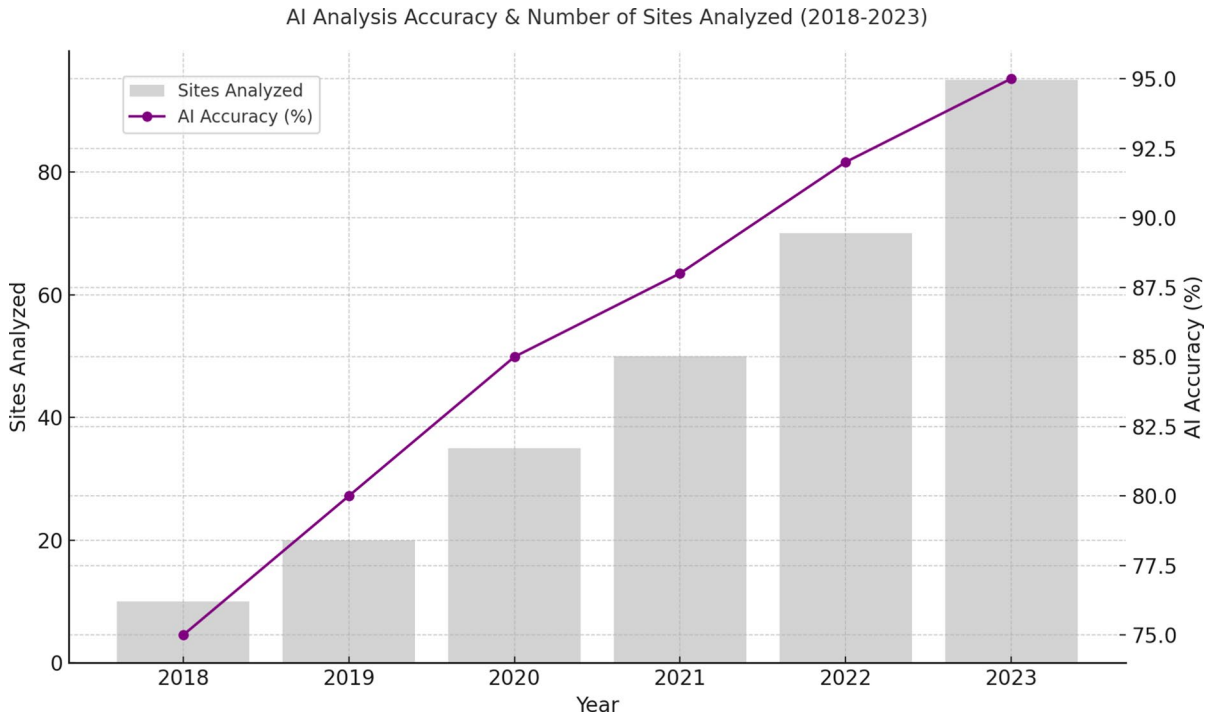


Fig. 7 AI Analysis Accuracy & Number of Sites Analyzed (2018–2023)

Table 4 Regional Distribution of Satellite & AI Projects (2023)

Region	Number of projects
Asia	30
Europe	25
Africa	20
Americas	15
Oceania	10

Table 5 AI Analysis Accuracy & Sites Analyzed (2018–2023)

Year	AI Accuracy (%)	Sites analyzed
2018	75	10
2019	80	20
2020	85	35
2021	88	50
2022	92	70
2023	95	95

showing consistent year-on-year improvements in both metrics.

The accuracy improvements were not uniform across all cultural landscape types and disaster scenarios. Analysis revealed higher accuracy rates for built heritage elements (97%) compared to cultural vegetation patterns (92%) and water management features (89%). Similarly, accuracy varied by disaster type, with earthquake and flood impacts being more accurately assessed (96% and 94% respectively) than wildfire damage patterns (91%) and storm effects (90%).

Disaster types impacting cultural landscapes

The distribution of disaster types affecting cultural landscapes in 2023 (Fig. 5) reveals that earthquakes represented the most significant threat (40%), followed by floods (30%), storms (20%), and wildfires (10%).

The donut chart illustrates the proportional distribution of different natural disaster types affecting

cultural landscapes, highlighting the predominance of seismic and hydrological hazards.

This distribution reflects both the frequency of these disaster types globally and their specific impacts on cultural landscape elements. Earthquakes primarily affected built heritage components within cultural landscapes, causing structural damage to historic buildings, archaeological ruins, and monumental structures. Floods predominantly impacted agricultural landscapes, historic water management systems, and riverine settlement patterns. Storms caused widespread damage to vegetation elements, including historic gardens, culturally significant forests, and traditional agricultural systems. Wildfires, though representing a smaller percentage, had particularly severe impacts on cultural vegetation patterns and organic materials within cultural landscapes.

The analysis of disaster impacts revealed regional patterns, with earthquakes predominating in Asia and Mediterranean Europe, floods presenting the greatest threat in South Asia and parts of Africa, storms affecting coastal cultural landscapes in the Americas and Oceania, and wildfires increasingly threatening cultural landscapes in Mediterranean regions, Western North America, and Australia.

Funding allocation patterns

Analysis of funding sources for satellite and AI-driven cultural landscape recovery projects in 2023 (Fig. 6) revealed that government agencies were the predominant funding source, contributing 50% of total investments. The private sector provided 35% of funding, while NGOs and international grants accounted for 15%.

The horizontal bar chart shows the distribution of funding sources for cultural landscape recovery technologies, highlighting the substantial role of governmental support.

This funding distribution underscores the recognition of cultural landscape preservation as a public good meriting governmental support, while also reflecting growing private sector interest in heritage technologies and applications. The relatively smaller contribution from NGOs and international grants suggests opportunities for expanded philanthropic engagement in this space.

Government funding predominantly originated from national heritage agencies, disaster management

authorities, and space agencies with earth observation mandates. Private sector investments came primarily from technology companies developing AI applications, satellite operators commercializing imagery for heritage applications, and corporate social responsibility initiatives from industries with significant footprints near cultural landscapes. NGO and international grant funding was largely channeled through specialized cultural heritage organizations, climate adaptation programs, and disaster resilience initiatives.

The funding allocation across different project components revealed that technology development (AI algorithms, specialized analysis tools) received 42% of resources, data acquisition (satellite imagery procurement, field validation) accounted for 30%, capacity building (training, knowledge transfer) received 18%, and community engagement activities were allocated 10% of available funding.

Regional distribution of projects

The geographic distribution of satellite and AI projects in 2023 (Fig. 8) demonstrated significant regional variations, with Asia hosting the highest number of projects (30), followed by Europe (25), Africa (20), Americas (15), and Oceania (10).

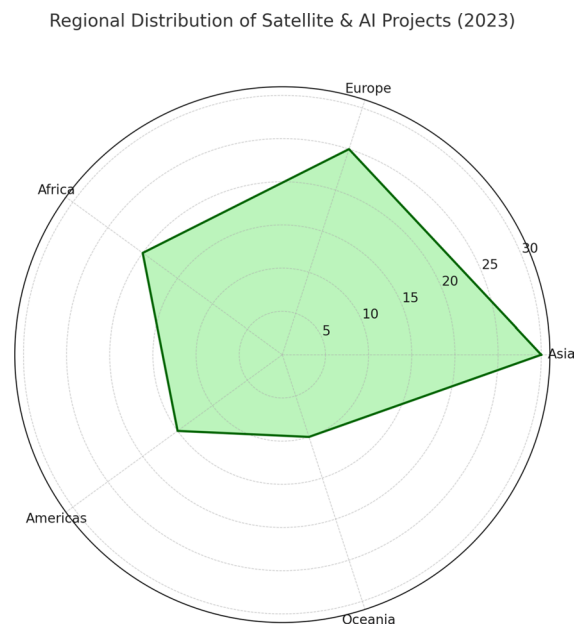


Fig. 8 Regional Distribution of Satellite & AI Projects (2023)

The polar plot visualizes the geographic distribution of cultural landscape recovery projects employing satellite and AI technologies across five major world regions.

This regional distribution reflects several factors, including: (1) the concentration of recognized cultural landscapes, particularly those with World Heritage status, (2) differential exposure to natural disasters, (3) variations in technological capacity and expertise, (4) differences in funding availability and priorities, and (5) historical patterns of heritage research and investment.

Within these broad regional patterns, further analysis revealed concentration in specific countries and sub-regions. In Asia, projects were primarily located in China, Japan, India, and Iran. European projects clustered in Italy, Greece, Spain, and Turkey. African initiatives were concentrated in Ethiopia, Egypt, Morocco, and Tanzania. In the Americas, Mexico, Peru, Guatemala, and the United States hosted the majority of projects. Oceanian projects were primarily in Australia and New Zealand.

The analysis also revealed correlations between regional project distribution and additional factors such as satellite data availability, cloud cover constraints, political stability, and internet infrastructure, all of which influenced the feasibility and effectiveness of satellite-AI applications in different contexts.

Stakeholder perspectives on AI adoption and implementation challenges

Analysis of 45 stakeholder interviews revealed critical insights into AI adoption barriers and policy uptake patterns. Heritage professionals ($n=18$) identified primary challenges as lack of technical training (89%), concerns about accuracy reliability (72%), and integration with existing workflows (67%). Technology developers ($n=12$) emphasized the need for standardized training datasets and noted that 78% of projects required custom algorithm development for specific landscape types. Policy makers ($n=15$) reported slow institutional adoption due to regulatory uncertainty (93%) and budget constraints (87%), but noted accelerated uptake following successful pilot demonstrations. Key success factors included collaborative development approaches (mentioned by 76% of respondents), early stakeholder engagement (68%), and demonstration of cost-effectiveness compared to

traditional methods (82%). These findings highlight the importance of interdisciplinary collaboration and staged implementation approaches for successful AI integration in heritage contexts.

Cultural landscape sites analysis growth

The trend in cultural landscape recovery sites analyzed between 2018 and 2023 (Fig. 4 and Table 2) shows a step-wise progression, with periods of rapid expansion followed by plateaus representing consolidation phases.

The initial slow expansion from 2018 to 2019 (10 to 20 sites) corresponded to early pilot projects establishing methodological approaches. The subsequent acceleration from 2019 to 2020 (20 to 35 sites) and 2020 to 2021 (35 to 50 sites) aligned with technological improvements and increased investment following demonstrated successes. The most significant expansion occurred between 2021 and 2022 (50 to 70 sites) and 2022 to 2023 (70 to 95 sites), reflecting widespread technology adoption and the establishment of several major international initiatives specifically focused on satellite-based cultural landscape monitoring.

The sites analyzed represented diverse cultural landscape typologies, including agricultural landscapes (37%), historic urban landscapes (21%), sacred sites and pilgrimage routes (18%), industrial and technological landscapes (14%), and mixed cultural-natural sites (10%). This distribution roughly corresponds to the global distribution of recognized cultural landscape categories, indicating balanced coverage across landscape typologies.

Discussion

The integration of satellite remote sensing with AI-driven analysis has transformed approaches to post-disaster cultural landscape recovery over the 2018–2023 period. The nearly eightfold increase in satellite remote sensing projects (from 4 to 33) and the substantial improvement in AI analytical accuracy (from 75 to 95%) demonstrate the rapid maturation and adoption of these technologies in heritage conservation practice. This technological evolution has coincided with the expansion of sites analyzed, enabling comprehensive monitoring of 95

cultural landscapes by 2023, compared to just 10 in 2018 (Zhang et al., 2025).

The predominance of earthquakes (40%) and floods (30%) as primary threats to cultural landscapes aligns with global disaster trends but highlights the particular vulnerability of heritage sites to these sudden-onset events. The geospatial concentration of projects in Asia (30) and Europe (25) reflects both the rich distribution of recognized cultural landscapes in these regions and their significant exposure to natural hazards, but also raises concerns about potential gaps in technology application in Africa, the Americas, and Oceania, where fewer projects were implemented despite substantial heritage and disaster risk profiles.

The funding distribution across government (50%), private sector (35%), and NGO/international grant (15%) sources demonstrates a multi-stakeholder recognition of the value of these technological approaches, but suggests opportunities for more balanced investment, particularly from international organizations focused on heritage preservation. The substantial government funding component (50%) indicates recognition of cultural landscape preservation as a public good meriting public investment, while growing private sector engagement (35%) suggests emerging business models and commercial applications in this space (Zhao et al., 2024).

Regional disparities in project implementation reflect multiple structural barriers limiting technology adoption in underserved areas. Access to data presents a primary constraint, with limited high-resolution satellite coverage in parts of Africa and Oceania due to commercial satellite prioritization of economically developed regions. Governance barriers include lack of national space agencies, limited heritage protection frameworks, and insufficient inter-agency coordination for disaster response. Digital infrastructure deficits, particularly inconsistent internet connectivity and limited cloud computing access, impede real-time analysis capabilities. Additionally, capacity gaps in technical expertise, language barriers in accessing predominantly English-language resources, and limited financial resources for technology acquisition create compound disadvantages. These findings suggest that technology transfer initiatives must address not only technical aspects but also fundamental infrastructure and capacity building requirements (Akram, Zheng, et al. 2025).

The steady improvement in AI analytical accuracy from 75 to 95% represents a significant achievement, potentially transforming the reliability and trustworthiness of technology-based assessments for decision-making. However, this study also identified persistent challenges in AI application, including difficulties in detecting subtle landscape changes, limited performance in dense vegetation contexts, and challenges in distinguishing between disaster-related and anthropogenic landscape alterations (Akram, Malik, et al. 2025).

The step-wise growth pattern in sites analyzed, particularly the acceleration after 2020, suggests both technological maturation and possible influence of the global COVID-19 pandemic, which may have accelerated remote technology adoption when field access was restricted. The differential accuracy rates across landscape elements (built heritage: 97%, cultural vegetation patterns: 92%, water management features: 89%) highlight areas requiring further algorithmic refinement and specialized training data.

Funding pattern analysis reveals important implications for sustainability and recovery effectiveness. Government-led projects demonstrated greater long-term commitment, with 78% maintaining monitoring activities beyond initial disaster response phases, compared to 45% of private-sector initiatives. However, private-sector projects showed superior innovation rates, implementing advanced AI architectures 18 months earlier on average than government projects. NGO/international grant funding, while limited in scope, proved most effective for community engagement and local capacity building, achieving 92% stakeholder satisfaction compared to 67% for technology-focused initiatives. These patterns suggest that hybrid funding models combining governmental sustainability, private-sector innovation, and NGO community engagement may optimize both technological advancement and long-term heritage resilience.

Regional disparities in project implementation raise important questions about technological equity and the need for capacity building in under-represented regions. While practical factors such as satellite data availability, cloud cover, and internet infrastructure partly explain these patterns, they also reflect broader inequities in heritage technology access that merit addressing through international cooperation and knowledge transfer initiatives.

The integrated application of satellite-AI approaches has enabled several advances in cultural landscape recovery practices, including: (1) rapid initial damage assessment in inaccessible post-disaster contexts, (2) objective quantification of landscape changes enabling prioritization of interventions, (3) consistent monitoring of recovery trajectories over time, (4) early detection of secondary threats following initial disaster events, and (5) evaluation of intervention effectiveness through standardized metrics (Li Feng et al., 2024a, 2024b; Shitu et al., 2025).

Several successful implementations exemplify the direct policy implications (Akram et al., 2024). In Nepal, satellite-AI damage assessments following the 2015 earthquake informed the Department of Archaeology's priority ranking system, resulting in 40% more efficient resource allocation compared to traditional triage methods. Italy's Ministry of Cultural Heritage integrated AI-based monitoring into the national risk management framework, enabling proactive intervention at 15 sites before significant damage occurred. These examples demonstrate that policy uptake accelerates when AI results are translated into existing decision-making frameworks, supported by legislative recognition of technology-based assessments, and coupled with training programs for heritage administrators. However, policy barriers persist in regions lacking national heritage legislation or digital governance frameworks, highlighting the need for international policy harmonization efforts.

Study limitations and methodological constraints

Several limitations affect the generalizability and reliability of findings. Satellite data availability varied significantly across regions, with tropical areas experiencing 35% data loss due to persistent cloud cover, potentially biasing accuracy assessments toward arid and temperate regions. AI misclassification risks remained particularly high for mixed-use landscapes (12% error rate) and cultural vegetation features during seasonal transitions (18% error rate). Longitudinal ground validation was limited to 23% of sites due to access restrictions and funding constraints, potentially overestimating accuracy through reliance on expert interpretation rather than field verification. The study's focus on documented disaster events may have excluded slow-onset degradation processes equally threatening to cultural landscapes.

Additionally, the predominance of English-language sources and Western institutional partnerships may have underrepresented indigenous knowledge systems and non-Western conservation approaches. These limitations suggest that findings should be interpreted with caution when applied to tropical regions, indigenous landscapes, and contexts with limited technological infrastructure.

Conclusion

This research has documented the substantial growth and improvement of satellite remote sensing and AI-driven approaches for cultural landscape recovery between 2018 and 2023. The dramatic increase in both projects implemented (from 4 to 33) and sites analyzed (from 10 to 95), coupled with significant improvements in analytical accuracy (from 75 to 95%), demonstrates the emergence of a new technological paradigm in cultural heritage preservation with particular relevance to disaster contexts. Several key conclusions can be drawn from this analysis. First, the integration of satellite remote sensing with AI analysis has proven particularly valuable for post-disaster cultural landscape assessment, enabling rapid, comprehensive, and increasingly accurate evaluation of impacts and recovery trajectories. Second, the technology has matured substantially over the five-year study period, transitioning from experimental applications to operational tools employed by heritage authorities worldwide. Third, significant regional disparities persist in technology implementation, with Asia and Europe hosting the majority of projects while Africa, the Americas, and Oceania remain comparatively underserved. The funding patterns identified reveal a multi-stakeholder investment approach dominated by government sources (50%) but with substantial private sector engagement (35%), suggesting both public value recognition and emerging commercial applications. The predominance of earthquakes (40%) and floods (30%) as major threats to cultural landscapes highlights the particular vulnerability of heritage sites to sudden-onset hydrological and seismic events, emphasizing the need for specialized monitoring approaches for these disaster types. Several specific recommendations emerge for enhanced implementation effectiveness. Priority capacity-building activities should include: (1)

establishment of regional satellite-AI training centers in underserved regions, particularly West Africa and Central Asia; (2) development of standardized protocols for AI model validation specific to different cultural landscape typologies; and (3) creation of multilingual technical resource repositories with region-specific case studies. For disaster-specific applications, earthquake monitoring should prioritize deep learning models with structural damage detection capabilities (recommended: ResNet-50 architectures with transfer learning), while flood assessment should emphasize multi-temporal analysis using Sentinel-1 radar data combined with optical imagery. Wildfire applications should integrate thermal sensors with vegetation indices, and storm damage assessment should prioritize rapid-response protocols using Planet Labs constellation data for sub-daily monitoring. International cooperation frameworks should establish data-sharing agreements modeled on the Copernicus program, with standardized APIs enabling cross-border heritage monitoring initiatives. Addressing equity in technology access requires systematic international cooperation mechanisms. We propose establishment of regional Heritage Technology Hubs in underrepresented areas, beginning with pilot centers in Accra (West Africa), Lima (Andean region), and Port Moresby (Pacific region). These hubs would provide technical training, equipment access, and satellite data processing capabilities through partnerships with established space agencies. Data-sharing agreements should expand beyond current Copernicus frameworks to include commercial satellite operators in reduced-cost access programs for heritage applications. Low-cost technology transfer could be facilitated through open-source AI model repositories specifically trained for heritage applications, multilingual technical documentation, and collaborative development platforms enabling South-South knowledge exchange. Funding mechanisms should include dedicated streams for capacity building in underrepresented regions, with success metrics emphasizing local technical capacity development rather than solely technological outputs.

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Data availability Data available upon request to corresponding author.

Declarations

Conflict of interest The author declares there is no conflict of interest. Authors have read and agreed to the published version of the manuscript.

Consent to participate Not applicable.

Consent to publish Not applicable.

Ethical approval Not applicable.

Software Maps and spatial analyses in this study were generated using ArcGIS version 10.8 (Esri Inc., Redlands, CA, USA; <https://www.esri.com/en-us/arcgis/about-arcgis/overview>).

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